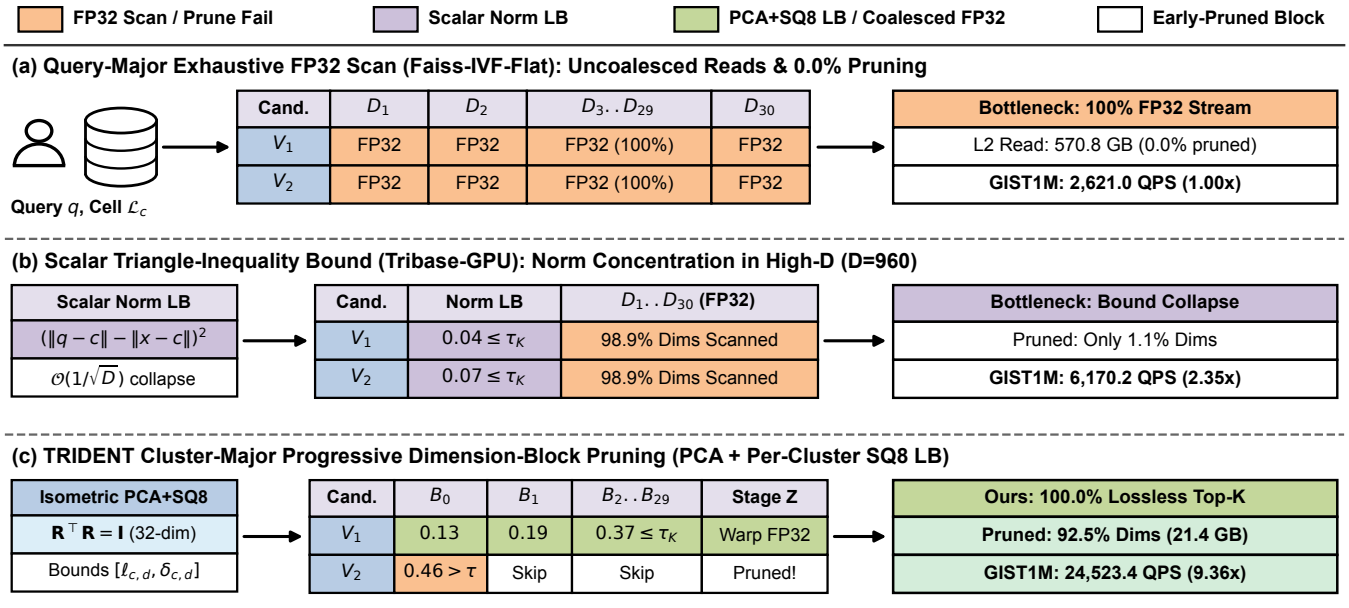

FP32 Scan / Prune Fail
Scalar Norm LB
PCA+SQ8 LB / Coalesced FP32
Early-Pruned Block
(a) Query-Major Exhaustive FP32 Scan (Faiss-IVF-Flat): Uncoalesced Reads & 0.0% Pruning
Query q, Cell $\mathcal{L}_c$
Cand. D_1 D_2 $D_3..D_{29}$ D_{30}
V_1 FP32 FP32 FP32 (100%) FP32
V_2 FP32 FP32 FP32 (100%) FP32
Bottleneck: 100% FP32 Stream
L2 Read: 570.8 GB (0.0% pruned)
GIST1M: 2,621.0 QPS (1.00x)
(b) Scalar Triangle-Inequality Bound (Tribase-GPU): Norm Concentration in High-D (D=960)
Scalar Norm LB
$(\|q-c\|-\|x-c\|)^2$
$\mathcal{O}(1/\sqrt{D})$ collapse
Cand. Norm LB $D_1..D_{30}$ (FP32)
V_1 $0.04 \le \tau_K$ 98.9% Dims Scanned
V_2 $0.07 \le \tau_K$ 98.9% Dims Scanned
Bottleneck: Bound Collapse
Pruned: Only 1.1% Dims
GIST1M: 6,170.2 QPS (2.35x)
(c) TRIDENT Cluster-Major Progressive Dimension-Block Pruning (PCA + Per-Cluster SQ8 LB)
Isometric PCA+SQ8
$\mathbf{R}^\top\mathbf{R}=\mathbf{I}$ (32-dim)
Bounds $[\ell_{c,d}, \delta_{c,d}]$
Cand. B_0 B_1 $B_2..B_{29}$ Stage Z
V_1 0.13 0.19 $0.37 \le \tau_K$ Warp FP32
V_2 $0.46 > \tau$ Skip Skip Pruned!
Ours: 100.0% Lossless Top-K
Pruned: 92.5% Dims (21.4 GB)
GIST1M: 24,523.4 QPS (9.36x)